

AGI: The Eternal Path

From Induction to Deduction, Towards True Wisdom

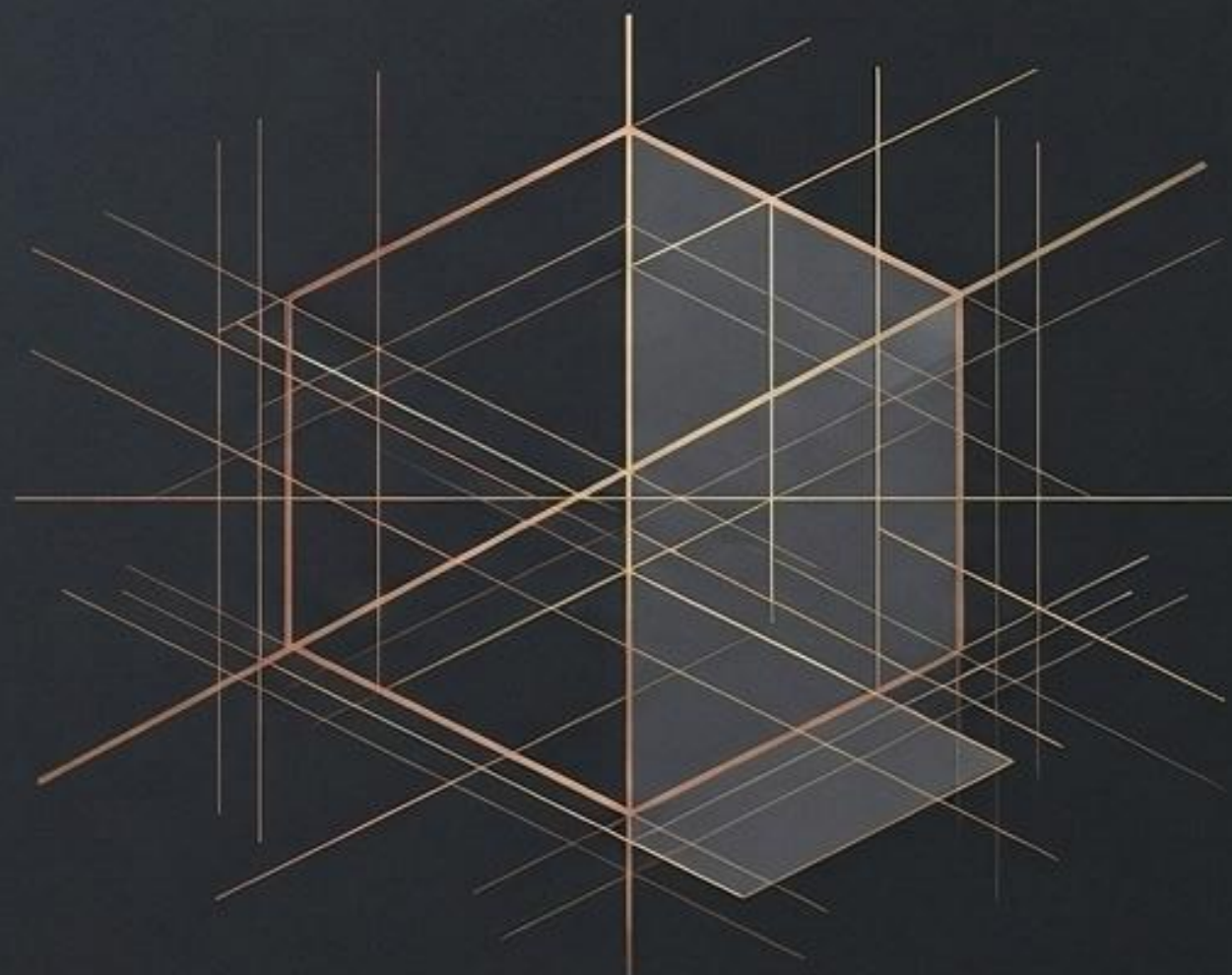


Type: System Architecture and
Philosophical Discourse

The Limits of Large Models: We Have Only Touched Half of Intelligence

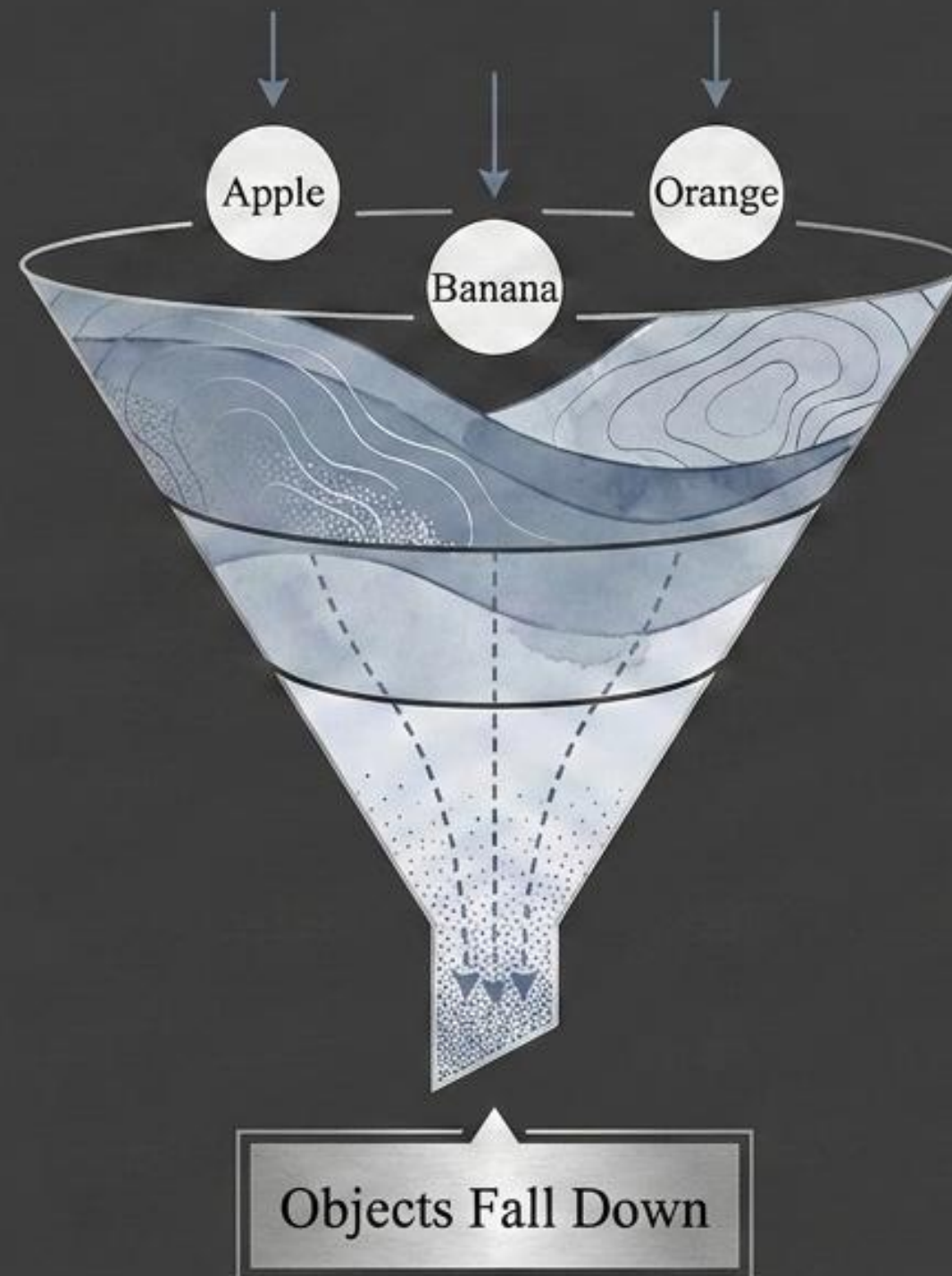


Large Language Models (LLMs) have demonstrated incredible generative capabilities. Writing articles, drawing, translating, programming, simulating conversations.



However, this is only half of intelligence. Large models excel at '**Induction**', but are not truly proficient at '**Deduction**'. This gap is the key watershed between current AI and true AGI (Artificial General Intelligence).

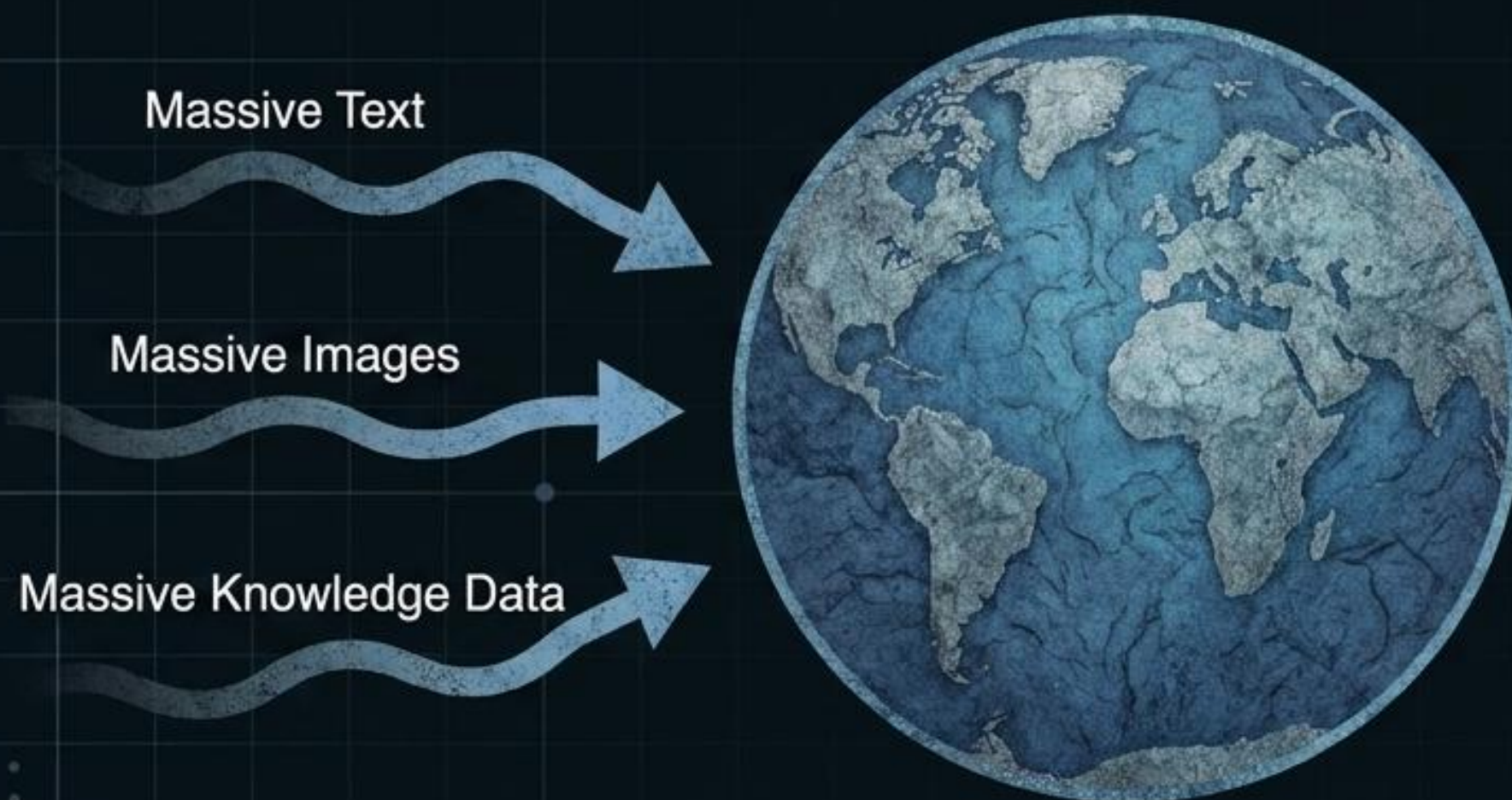
Inductive Reasoning: From Massive Observations to a Single Rule



The core of Induction:
Identifying common patterns
from numerous independent
results.

Inductive Engine: A Statistical Machine Compressing the World

Based on Slide C, the content is translated into English



Large Language Models are essentially:
Super-Large Statistical Induction Systems.

Powerful Generation Capability
Articles, Images, Code

Powerful Pattern Recognition
Discovering Hidden Patterns and Trends

Powerful Knowledge Compression Capability
Condensing Billions of Data Points into
Parameters

The Illusion of Meaning: Co-occurrence $\neq$ Causality



LLMs build 'co-occurrence relationships' between words, sentences, and images.

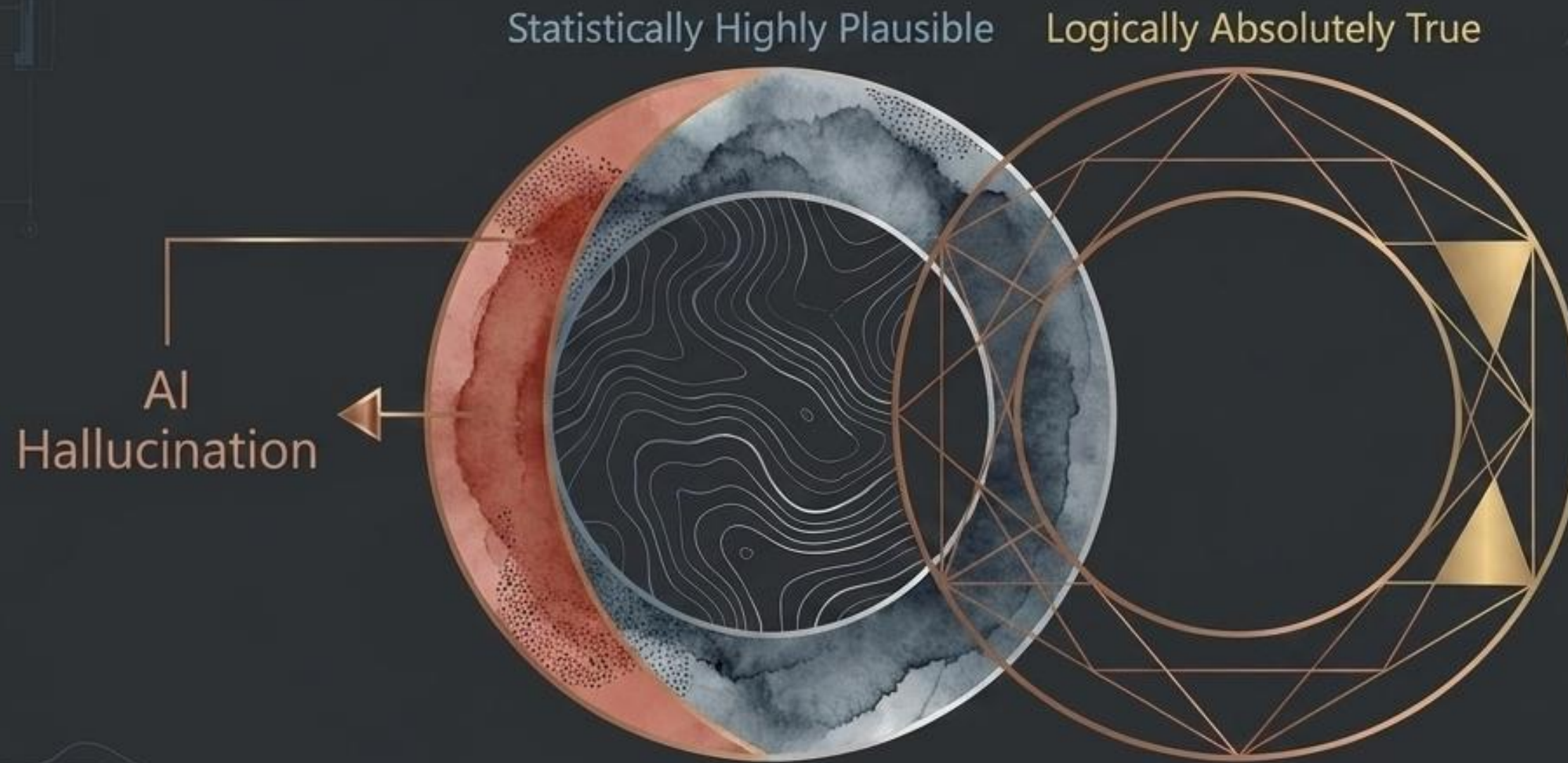
These are just 'CORRELATIONS', not 'CAUSALITY'.

Statistical Trap: When Correlation Masquerades as Causation



This is the trap of induction: Seeing only correlation, missing the truth.

The Anatomy of Hallucination: The Disconnect Between Probability and Logic



Models predict the most probable answer, rather than verifying if the answer is true.

- Non-existent Papers
- Non-existent People
- Incorrect Citations

The Fatal Blind Spot: High-Risk Fields Cannot Afford "Possibilities"



Medical

Finds statistical correlations in symptoms, but fails to find the true cause of the disease.



Finance

Discovers market fluctuation patterns, but ignores the underlying causal mechanisms.



Industrial Safety

Sees historical statistics of accidents, but cannot deduce the root cause of the accident.

Conclusion: High-reliability systems cannot rely on induction alone.

The Sword of Deduction: Precise, Rigorous, and Verifiable Logic

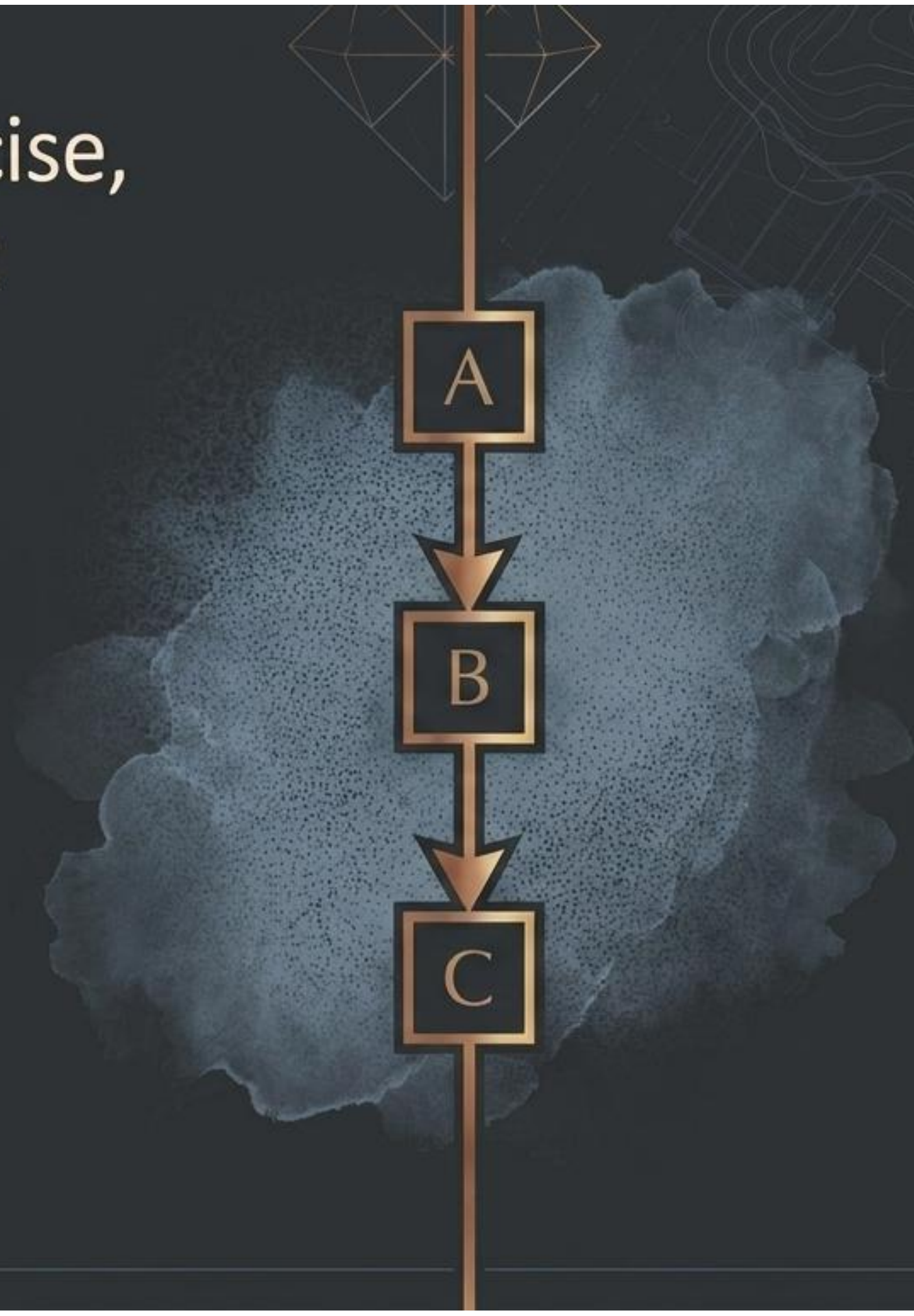
Deduction is like a sharp sword.

Characteristics:

Precise, rigorous, absolutely verifiable.

Strengths:

- Logical reasoning
- Answer verification
- Key decision-making and thinking methods



Yin and Yang: The Inclusion of the Sea and the Precision of the Sword

Yin - Induction



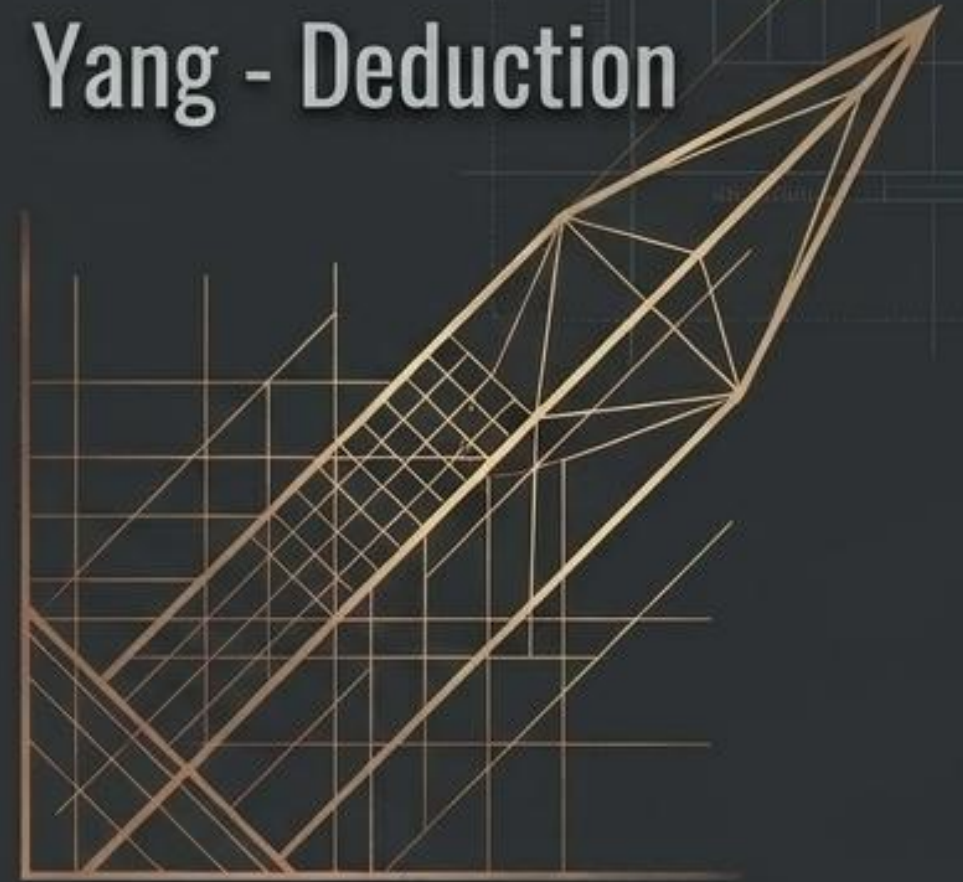
Like the ocean.

Inclusive of all, absorbs information.

Good at generation, learning, association.

Knowledge without guarantee.

Yang - Deduction



Like a sharp sword.

Precise and rigorous, resolves doubts and confusion.

Good at reasoning, verification, decision-making.

Logic without knowledge.

AGI System Diagnostic Matrix: Yin-Yang Complementary Framework

Induction (Yin)	Deduction (Yang)	Taiji Unity (AGI)
Static 	Dynamic 	Static and Dynamic Coexistence, Complementary 
Massive Data 	Few Premises 	Massive Data + Precise Premises 
Effect ↔ Effect (Statistical Occurrence) 	Cause ↔ Effect (Transport Reasoning) 	Causation + Co-occurrence 
Imprecise (Black Box) 	Precise (White Box) 	Model Flexibility + Rigorous Variation 
For Reference 	For Industry 	Indrade Reliable Relibration 
Common Features 	Associative Reasoning 	Generation + Reasoning 
Mistaking Correlation for Causation 	Insufficiencies Lead to Errors 	Centrality Reduce Risk 
Strong Generation 	Strong Reasoning 	AGI Eternal Way 

Three-fold Fusion Loop: Engineering Realization of the Tao

Ring 1

Data-Driven → Rule Generation

Induction: Identify patterns from big data.
Deduction: Transform patterns into interpretable rules.

Ring 3

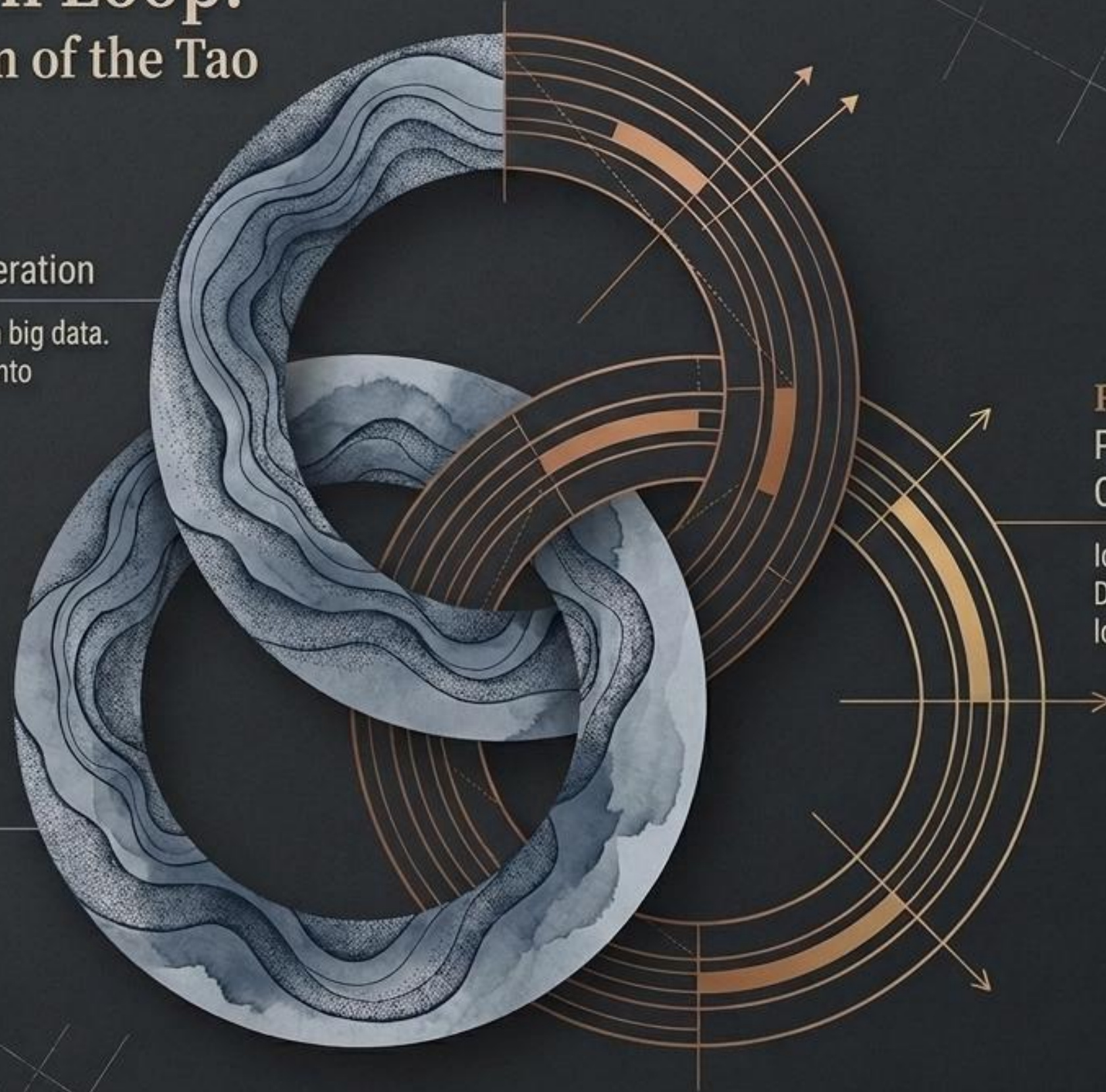
Fuzzy → Precise

Provide general directions and possibilities.
Deduction: Provide rigorous verification and judgment.

Ring 2

Phenomenon Association → Causal Reasoning

Identify 'Effect ↔ Effect' co-occurrence.
Deduction: Establish 'Cause → Effect' logical chains.



The Eternal Wisdom Formula

AGI = Induction (Statistics) + Deduction (Reasoning)

Wisdom = Experience x Logic

Reliability = Correlation + Causality

The Eternal Tao = Yin and Yang in Harmony. Only through the union of Yin and Yang will AI no longer just imitate, but truly understand; no longer just generate, but verify; becoming a truly reliable wisdom partner.

About Us



LonGeat

It is a team focused on AGI (Artificial Intelligence General) research and development practices.



Typeology

New knowledge

Mind Model

Knowledge Unified Model

Mind AI

AGI Core Platform

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